Algebraic rootfinding methods: a conceptual survey

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Abstract

Algebraic rootfinders represent the finite zero set of a polynomial system by a matrix, a matrix polynomial, or a univariate polynomial. We explain seven constructions: Sylvester and Cayley–Bézout resultants, Macaulay matrices, multiparameter eigenvalue problems, Gröbner elimination, rational univariate representations, and normal-form methods. A common quotient-algebra framework explains why these representations work, while a single four-root example makes their differences explicit. Short proofs cover the main algebraic identities; more substantial existence and regularity results are stated with their hypotheses and references. We distinguish exact root correspondence from numerical stability, including the effects of projection, basis choice, and the norm in which data are perturbed. The survey provides a methodological foundation for studying conditioning without presupposing an average-case verdict on the methods.

1 The problem and a guide to the methods

Let $P = (p_1, \dots, p_d)$ be a polynomial map $\mathbb{C}^d \to \mathbb{C}^d$. We assume throughout that $I = \langle p_1, \dots, p_d \rangle$ is radical and zero-dimensional, with distinct roots $z_1, \dots, z_N$, and that $J(z_j) = DP(z_j)$ is invertible. These assumptions exclude multiple roots and positive-dimensional solution sets. They make the central ideas transparent; methods for multiplicities require additional local-algebra machinery [27, 36].

The task is to compute all N roots. Newton iteration is local, whereas homotopy continuation can find all roots by tracking a suitable collection of paths. The methods considered here organize the computation differently:

coefficients $\longrightarrow$ algebraic representation $\longrightarrow$ spectral or scalar roots $\longrightarrow$ solutions of P.

We will ask three questions of each construction: what is built, why does it encode the roots, and how are all coordinates recovered? The answers also identify three places where numerical errors can enter: construction, solution of the intermediate problem, and recovery. Exact equivalence of representations does not imply equal conditioning of these operations.

1.1 Three algorithmic families

Resultant and eigenvalue constructions.
Build a matrix that becomes singular at a root or at a projected root. Sylvester and Cayley–Bézout methods hide one variable; Macaulay methods adjoin a linear form; multiparameter methods replace the equations by affine matrix pencils.

Scalar elimination.
Build a univariate polynomial whose roots label the solutions, together with formulas recovering their coordinates. Gröbner elimination and rational univariate representations take this route.

Normal forms.
Construct multiplication operators on the finite-dimensional quotient algebra $A = \mathbb{C}[x_1, \ldots, x_d]/I$, then recover the roots from their common eigenvectors.

These families overlap. A scalar eliminant can be a characteristic polynomial of a multiplication matrix, and a Macaulay matrix can be used to construct a normal form. The classification describes how the representation is computed, not mutually exclusive algebraic objects. Table 1 at the end collects their sizes and hypotheses.

Example 1.1 (A running system)**.** We will follow the same system through the constructions:

$$p(x, y) = x^2 - 1 = 0, \qquad q(x, y) = y^2 - x = 0. \tag{1}$$

Its roots are

$$(1, 1), \quad (1, -1), \quad (-1, i), \quad (-1, -i).$$

Indeed $x = y^2$ and $y^4 = 1$. The Jacobian determinant is $4xy$, which is nonzero at all four roots. The coordinate y distinguishes the roots, while x does not: there are two roots above each value of x. This distinction will explain why a simple polynomial root need not give a simple projected eigenvalue.

1.2 Degree conventions and roots at infinity

For total degrees $e_1, \ldots, e_d$, a generic system has $\prod_i e_i$ projective roots by Bézout's theorem, and all are affine if there is no common root on the hyperplane at infinity. If instead every equation has degree at most n in *each* variable, its support lies in the box $[0, n]^d$. The generic count is $d!n^d$, naturally computed in $(\mathbb{P}^1)^d$; see the multihomogeneous and sparse resultant accounts in [9, 16]. The running example has total degrees $(2, 2)$ and four roots; it is not a generic box-degree-two system, which would have eight.

The compactification matters. Ordinary homogenization of a full box-degree-n equation to total degree dn has top part $c(x_1 \cdots x_d)^n$. For $d \geq 2$, the resulting equations have common zeros at ordinary projective infinity even when the system is generic in its box support. Generic absence of boundary zeros in $(\mathbb{P}^1)^d$ does not remove these ordinary projective zeros. We therefore state the no-infinity assumption separately for the dense Macaulay construction.

2 The common algebra: functions on a finite root set

Write $R = \mathbb{C}[x_1, \dots, x_d]$ and $A = R/I$. The class $[f] \in A$ records the values of f at the roots; adding a polynomial in I changes none of them. The following elementary form of the Chinese remainder theorem is the organizing fact.

Proposition 2.1 (Evaluation and interpolation)**.** *Evaluation is an algebra isomorphism*

$$\mathrm{ev} : A \longrightarrow \mathbb{C}^N, \qquad [f] \longmapsto (f(z_1), \dots, f(z_N)),$$

where multiplication on $\mathbb{C}^N$ is componentwise. In particular $\dim A = N$*. There are unique classes $\ell_j \in A$ with $\ell_j(z_k) = \delta_{jk}$; they satisfy $\ell_j \ell_k = \delta_{jk}\ell_j$ and $\sum_j \ell_j = 1$.*

Proof. Because I is radical, a polynomial vanishes at every root exactly when it belongs to I (Hilbert's Nullstellensatz), so evaluation is injective. Choose a linear form t taking distinct values $t_j = t(z_j)$. Such choices exist: the prohibited coefficient vectors lie in the finite union of hyperplanes $\{u : u^T(z_j - z_k) = 0\}$, $j \neq k$. The polynomials

$$\ell_j(x) = \prod_{k \neq j} \frac{t(x) - t_k}{t_j - t_k}$$

evaluate to the coordinate vectors of $\mathbb{C}^N$, proving surjectivity. The remaining identities follow by evaluation and injectivity. □

A form taking distinct values on the root set is called *separating*. The proof is conceptual: algorithms compute the quotient from the coefficients; they do not first compute the roots to construct the ℓ_j.

2.1 Multiplication becomes diagonal

For $g \in R$, multiplication by $[g]$ is a linear operator on A. In a basis $\mathcal{B} = (b_1, \dots, b_N)$, write its matrix as M_g, with column k equal to $[gb_k]_{\mathcal{B}}$. Define the evaluation matrix

$$V = (b_k(z_j))_{k,j}, \qquad v_j = \mathcal{B}(z_j).$$

Columns index roots; coefficient vectors are columns. Throughout, ^{T} denotes transpose and * conjugate transpose.

Proposition 2.2 (The multiplication eigenvalue theorem)**.** *The matrix V is invertible and*

$$M_g = V^{-T} \operatorname{diag}(g(z_1), \dots, g(z_N)) V^T. \tag{2}$$

Thus all multiplication matrices commute and are simultaneously diagonalizable. Their characteristic and minimal polynomials are, respectively,

$$\det(TI - M_g) = \prod_{j=1}^{N} (T - g(z_j)), \qquad \prod_{\eta \in \{g(z_1), \dots, g(z_N)\}} (T - \eta).$$

Proof. Evaluation sends the coefficient vector c to $V^T c$ and sends multiplication by g to multiplication by $\operatorname{diag}(g(z_j))$. Proposition 2.1 gives the invertibility of V and the asserted similarity. The polynomial identities follow for a diagonal matrix. □

In this convention the right eigenvector at a separated value $g(z_j)$ is $[\ell_j]_{\mathcal{B}} = V^{-T} e_j$, and v_j^T is a left eigenvector. Equivalently, the evaluation vectors are right eigenvectors of M_g^T. Repeated values of g create higher-dimensional eigenspaces even when every root of P is simple.

2.2 The univariate model

For a monic $p(T) = T^m + a_{m-1}T^{m-1} + \cdots + a_0$, reduction modulo p in the basis $(1, T, \ldots, T^{m-1})$ gives the companion matrix

$$C_p = \begin{pmatrix} 0 & & & -a_0 \\ 1 & 0 & & -a_1 \\ & \ddots & \ddots & \vdots \\ & & 1 & -a_{m-1} \end{pmatrix}, \qquad \det(TI - C_p) = p(T). \tag{3}$$

For squarefree p the determinant identity follows from Proposition 2.2; for every monic p it follows because $e_1, C_p e_1, \ldots, C_p^{m-1} e_1$ is a basis and $p(C_p) = 0$, so the minimal and characteristic polynomials both equal p. At a root λ, $(1, \lambda, \ldots, \lambda^{m-1})$ is a *left* eigenvector.

This is the basic representation change: a scalar polynomial equation becomes a matrix eigenproblem. The multivariate methods make different choices about which part of this pattern to preserve.

3 Family I: resultant and eigenvalue constructions

Here singular matrices detect roots. A *regular* square matrix polynomial $R(\lambda)$ has determinant not identically zero. Its finite eigenvalues are the zeros of that determinant; a singular matrix polynomial requires a different analysis. Even in the regular case, degree drops can produce eigenvalues at infinity. The following constructions specify when finite eigenvalues represent affine roots.

3.1 The Sylvester resultant ($d = 2$)

Idea. Freeze one coordinate, say y, and ask whether the resulting polynomials in x have a common root. The Sylvester matrix turns this question into singularity of a matrix whose entries are polynomials in y.

Construction and proof. First take nonzero univariate polynomials $f(x) = a_m x^m + \cdots + a_0$ and $g(x) = b_k x^k + \cdots + b_0$ of *actual* degrees $m, k \geq 1$. Write the coefficient vectors of

$$x^{k-1} f, \ldots, f, \quad x^{m-1} g, \ldots, g$$

as rows, with columns ordered $x^{m+k-1}, \dots, 1$. This defines the $(m+k) \times (m+k)$ Sylvester matrix $S(f,g)$. Its transpose represents the map

$$\mathbb{C}[x]_{<k} \oplus \mathbb{C}[x]_{<m} \longrightarrow \mathbb{C}[x]_{<m+k}, \qquad (u,v) \longmapsto uf + vg.$$

If f and g are coprime, $uf = -vg$ implies that f divides v; the degree bound forces $v = 0$, then $u = 0$. Conversely, if $h = \gcd(f,g)$ has positive degree, $(u,v) = (g/h, -f/h)$ is a nonzero vector in this kernel. Thus

$$S(f,g) \text{ is singular} \quad \Longleftrightarrow \quad f, g \text{ have a common root in } \mathbb{C}.$$

With this ordering its determinant is the resultant, with product formula

$$\det S(f,g) = \operatorname{Res}_x(f,g) = a_m^k \prod_{f(\alpha)=0} g(\alpha),$$

where roots are counted with multiplicity; see [9, Ch. 3]. The same argument shows $\dim \ker S(f,g) = \deg \gcd(f,g)$: every relation is $(u,v) = w(g/h, -f/h)$ with $\deg w < \deg h$.

For $p, q \in \mathbb{C}[x,y]$, choose their degrees m, k in x over $\mathbb{C}[y]$ and form $S(y)$ by the same shifts. Its determinant is a polynomial in y. At a parameter value where both degrees are preserved, the preceding proof applies verbatim. If (x_*, y_*) is an affine common root, direct evaluation of the rows gives

$$S(y_*) \begin{pmatrix} x_*^{m+k-1} & \cdots & x_* & 1 \end{pmatrix}^T = 0.$$

When the specialized gcd is linear, this vector spans the right kernel, so its last two entries recover x_* by a ratio. Several roots in the same fiber require additional work, such as computing the fiber gcd.

Example 3.1 (The running system)**.** For $p = x^2 - 1$ and $q = y^2 - x$ in Example 1.1, the degrees in x are 2 and 1. The rows p, xq, q give

$$S(y) = \begin{pmatrix} 1 & 0 & -1 \\ -1 & y^2 & 0 \\ 0 & -1 & y^2 \end{pmatrix}, \qquad \det S(y) = y^4 - 1.$$

At each $\eta \in \{1, -1, \mathrm{i}, -\mathrm{i}\}$, a right null vector is $(\eta^4, \eta^2, 1)^T = (1, \eta^2, 1)^T$. Thus $x = \eta^2$, recovering $(1, \pm 1)$ and $(-1, \pm \mathrm{i})$. All four y-values are distinct; hiding x instead would combine the roots into two fibers.

Remark 3.2 (A resultant remembers its degree convention). A matrix constructed with fixed degrees remains that size when leading coefficients vanish after specialization. It then tests common zeros of the *homogenized* polynomials in $\mathbb{P}^1$, including infinity. For example, $f = ax + 1$, $g = ax + 2$ give

$$S(a) = \begin{pmatrix} a & 1 \\ a & 2 \end{pmatrix}, \qquad \det S(a) = a.$$

At $a = 0$ the affine polynomials 1 and 2 have no root, but their degree-one homogenizations $aX + Z$, $aX + 2Z$ share $[X : Z] = [1 : 0]$. Consequently a vanishing fixed-degree resultant need not identify an affine solution. Degree drops and boundary roots must be checked when extending a projected candidate to a full solution.

Reduction and recovery. Solve the polynomial eigenvalue problem $S(y)v = 0$. For generic polynomials of degree n in each variable, S has size $2n$, degree n, and $2n^2$ finite eigenvalues, one per root. A companion linearization has size $2n^2$ and can be solved by QZ. Sizes and grades must be adapted to other supports: the running example has size 3 and degree 2, so its standard size-6 linearization also has two eigenvalues at infinity.

Numerical aspects and references. The entries of S depend linearly on the input coefficients. Nevertheless, arbitrary perturbations of S's coefficient matrices need not respect the repeated-coefficient structure of a Sylvester matrix. QZ controls backward error for its pencil; translating this into error in the original polynomials also requires control of the linearization and construction. See [42, 31] for these numerical issues. The classical sources are [38, 39]; accessible accounts include [9, Ch. 3] and [37].

3.2 The Cayley/Bézout resultant

Idea. A divided difference compresses the common-root test into a smaller matrix. Begin with two univariate polynomials q_1, q_2 of degree at most n. Their *Bézoutian* is

$$F(s,t) = \frac{q_1(s)q_2(t) - q_2(s)q_1(t)}{s-t} = v_n(s)^T B(q_1,q_2)v_n(t), \qquad v_n(t) = (1,t,\dots,t^{n-1})^T. \quad (4)$$

The numerator vanishes on $s = t$, so the quotient is a polynomial of degree at most $n-1$ in each variable. Interchanging s, t changes both numerator and denominator by a sign; hence $F(s,t) = F(t,s)$ and $B^T = B$. In particular, this is complex symmetry, not Hermitian symmetry. If ξ is a common root, then $F(s,\xi) = 0$ identically, and therefore $Bv_n(\xi) = 0$.

Proposition 3.3 (Bézout matrix and quotient multiplication)**.** *Suppose q_1 is monic of degree n, $\deg q_2 \le n$, and C_{q_1} is the subdiagonal companion matrix in* (3)*. Then*

$$B(q_1,q_2) = q_2(C_{q_1})B(q_1,1) = B(q_1,1)q_2(C_{q_1}^T). \quad (5)$$

Consequently

$$\det B(q_1,q_2) = (-1)^{n(n-1)/2} \prod_{q_1(\alpha)=0} q_2(\alpha) = (-1)^{n(n-1)/2} \operatorname{Res}(q_1,q_2).$$

Proof. Put $H(s,t) = (q_1(s) - q_1(t))/(s-t)$. The identity

$$F(s,t) - q_2(s)H(s,t) = q_1(s)\frac{q_2(t) - q_2(s)}{s-t}$$

shows that F is the remainder of $q_2(s)H$ modulo $q_1(s)$. Multiplication by $q_2(s)$ followed by this reduction acts on coefficient columns as $q_2(C_{q_1})$, proving the first equality. Transpose it

to obtain the second. The entries of $B(q_1, 1)$ on its anti-diagonal are 1, and those beyond it are zero, so its determinant is $(-1)^{n(n-1)/2}$. Finally, the eigenvalues of C_{q_1} are the roots of q_1, with multiplicity, giving the determinant formula. □

This is Barnett's factorization; [20] gives a broader account. Its order depends on the companion convention. If instead q_1 has degree n and leading coefficient $a \neq 0$, and q_2 has actual degree $k \leq n$, the formula becomes $\det B = (-1)^{n(n-1)/2} a^{n-k} \operatorname{Res}(q_1, q_2)$. The leading-coefficient factor therefore matters when the degrees differ. Bilinearity in the two sets of input coefficients follows directly from (4), rather than from powers of the companion matrix.

Example 3.4 (The same roots in a smaller matrix)**.** With $q_1(s) = s^2 - 1$, $q_2(s) = y^2 - s$, direct division gives

$$F(s,t) = -1 + y^2(s+t) - st, \qquad B(y) = \begin{pmatrix} -1 & y^2 \\ y^2 & -1 \end{pmatrix}, \qquad \det B(y) = 1 - y^4.$$

At $y = \eta$ with $\eta^4 = 1$, its kernel is spanned by $(1, \eta^2)^T$, which again gives $x = \eta^2$. The 2×2 quadratic matrix polynomial has a size-4 companion linearization, matching the four roots. The factorization is visible here: $C_{q_1} = B(q_1, 1) = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and $B(y) = (y^2 I - C_{q_1}) B(q_1, 1)$.

Several eliminated variables. Write $q_j = p_j[x_d]$, viewing x_d as a parameter, and introduce $s, t \in \mathbb{C}^{d-1}$. Define

$$F(s,t) = \frac{1}{\prod_{k=1}^{d-1}(s_k - t_k)} \det \begin{pmatrix} q_1(s_1, \ldots, s_{d-1}) & \cdots & q_d(s_1, \ldots, s_{d-1}) \\ q_1(t_1, s_2, \ldots, s_{d-1}) & \cdots & q_d(t_1, s_2, \ldots, s_{d-1}) \\ \vdots & & \vdots \\ q_1(t_1, \ldots, t_{d-1}) & \cdots & q_d(t_1, \ldots, t_{d-1}) \end{pmatrix}. \tag{6}$$

Successive rows agree when $s_k = t_k$, so every displayed factor divides the determinant and F is a polynomial. If each q_j has degree at most n in each variable, s_k occurs in the first k rows and t_k in the last $d - k$ rows. Division therefore gives

$$\deg_{s_k} F \leq kn - 1, \qquad \deg_{t_k} F \leq (d-k)n - 1.$$

Expand $F(s,t) = \sum_{\alpha,\beta} R_{\alpha\beta} s^\alpha t^\beta$ using these full exponent boxes. Both index sets have size $M = (d-1)! n^{d-1}$, producing a square matrix $R(x_d)$ of degree at most dn in the hidden variable. For $d > 2$, this matrix need not be symmetric.

If ξ is a common affine root of the specialized q_j, the last row in (6) vanishes at $t = \xi$. Cancelling the nonzero polynomial $\prod_k (s_k - \xi_k)$ proves $F(s, \xi) = 0$. Similarly $F(\xi, t) = 0$. Thus $(\xi^\beta)_\beta$ and $(\xi^\alpha)_\alpha$ are right and transpose-left null vectors. In a one-dimensional kernel they can be recovered up to scale. A coordinate is read from an available pair of monomials differing by that coordinate; coordinates absent from these vectors require back-substitution.

When is this an exact resultant? For the full common box support, the determinant is, up to a fixed nonzero normalization, the multihomogeneous resultant of the q_j on $(\mathbb{P}^1)^{d-1}$ [11,

Thm. 6.13]. This supplies the converse to the null-vector argument, including boundary roots in that compactification. For generic full-box p_j, the system has $d!n^d$ simple affine roots, no boundary roots in $(\mathbb{P}^1)^d$, and distinct hidden coordinates. Then $R(x_d)$ is regular and its finite eigenvalues are exactly those coordinates. Arbitrary sparse or total-degree inputs need a support-appropriate construction: padding the full-box matrix can make it identically singular.

Example 3.5 (A linear system and Cramer's rule)**.** Take $p_1 = x-z$, $p_2 = y-z$, $p_3 = x+y+z-1$ and hide z. Subtracting successive rows in (6) and dividing out the differences reduces its determinant to

$$\det \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ t_1 - z & t_2 - z & t_1 + t_2 + z - 1 \end{pmatrix} = 3z - 1.$$

The support-adapted Cayley matrix is the scalar $3z - 1$; back-substitution gives $x = y = z = 1/3$. Its coefficients are precisely the determinants in Cramer's rule. Padding this constant-in-s, t expression to the generic box size 2 would introduce a zero row and column.

Reduction and recovery. For generic full-box inputs, solve $R(x_d)v = 0$ of size $(d-1)!n^{d-1}$ and degree dn, for example through a size-$d!n^d$ linearization. Recover the other coordinates from null vectors or the original equations, checking boundary candidates for specialized inputs.

Numerical aspects and references. Each entry is homogeneous of degree d in the input coefficients, and linear in the coefficients of each polynomial separately. Scaling and the permitted perturbations therefore matter when comparing this eigenproblem with the original root problem; small matrix-coefficient errors are not automatically small input errors. The classical divided-difference constructions are [7, 12]; multihomogeneous determinantal formulae are treated in [11]. See [31] for the numerical analysis, including the general Cramer connection, and [30] for a bivariate implementation using subdivision and Chebyshev representations.

3.3 The Macaulay resultant and the u-resultant

Idea. Adjoin a linear form whose vanishing tests a chosen projection of the roots. If $F_i(x_0, x)$ is the homogenization of p_i and $h = \mu_0 x_0 + \sum_i \mu_i x_i$, the projective resultant $\mathrm{Res}(F_1, \ldots, F_d, h)$ vanishes precisely when a common root of the F_i lies on $h = 0$. With the F_i fixed, this polynomial in μ is the *u-resultant*. When all projective roots are simple and lie in $x_0 \neq 0$, its product formula is

$$\mathrm{Res}(F_1, \ldots, F_d, h) = c \prod_{j=1}^{N} h(1, z_j), \qquad c \neq 0.$$

Thus a separating linear form turns the root set into a univariate spectrum [43, 16].

A precise construction. Put $e_i = \deg p_i$, and assume that the F_i define a reduced, zero-dimensional projective complete intersection with no point on $x_0 = 0$. These are ordinary

projective hypotheses; they need not hold for a generic system with prescribed box support. Write

$$\widehat{R} = \mathbb{C}[x_0, \dots, x_d], \qquad \widehat{A} = \widehat{R}/\langle F_1, \dots, F_d \rangle, \qquad s = \sum_i (e_i - 1), \quad \rho = s + 1.$$

Here $\widehat{R}_t$ and $\widehat{A}_t$ denote the homogeneous degree-t pieces. The complete-intersection Hilbert series gives

$$\sum_{t \geq 0} (\dim \widehat{A}_t) T^t = \frac{\prod_i (1 + T + \cdots + T^{e_i - 1})}{1 - T}, \qquad \dim \widehat{A}_t = N = \prod_i e_i \quad (t \geq s).$$

Indeed, quotienting successively by a homogeneous nonzerodivisor of degree e_i multiplies the Hilbert series by $1 - T^{e_i}$; the displayed numerator has degree s and value N at $T = 1$. Evaluation at $(1, z_j)$ identifies $\widehat{A}_s$ and $\widehat{A}_\rho$ with $\mathbb{C}^N$; this is the standard Hilbert-function description of a reduced projective complete intersection [9, Chs. 3 and 5].

Let D_ρ have one column per degree-ρ monomial and coefficient rows for all mF_i with $\deg m = \rho - e_i$. Its row space is precisely the degree-ρ part of $\langle F_1, \dots, F_d \rangle$. Choose a matrix Z whose N columns form a basis of $\ker D_\rho$, and choose homogeneous lifts $b_1, \dots, b_N \in \widehat{R}_s$ of a basis of $\widehat{A}_s$. Both choices can be made by rank-revealing linear algebra on polynomial coefficient matrices. For a linear form h, form the $N \times N$ matrix

$$H_h = \begin{pmatrix} \operatorname{coeff}_\rho(b_1 h) \\ \vdots \\ \operatorname{coeff}_\rho(b_N h) \end{pmatrix} Z. \tag{7}$$

Coefficient vectors in this formula are rows, in the same monomial order as D_ρ. This compresses the h-rows after imposing all polynomial relations. Choose homogeneous linear forms α, β and put $h = \alpha - \lambda\beta$.

Proposition 3.6. *Under the stated assumptions,* $\det H_h = c' \prod_j h(1, z_j)$ *with* $c' \neq 0$ *independent of* h. *If* $\beta(1, z_j) \neq 0$ *for every* j, *then* H_β *is invertible and the pencil* $H_\alpha - \lambda H_\beta$ *has eigenvalues* $\lambda_j = \alpha(1, z_j)/\beta(1, z_j)$, *counted with multiplicity.*

Proof. Let V_ρ have as its jth column the degree-ρ monomials evaluated at $(1, z_j)$, and put $E = (b_i(1, z_j))_{i,j}$. Evaluation is an isomorphism in these degrees, so E is invertible and V_ρ has rank N. Since $D_\rho V_\rho = 0$, we have $Z = V_\rho C$ for some invertible C. Consequently

$$H_h = E \operatorname{diag}\Big(h(1, z_1), \dots, h(1, z_N)\Big) C.$$

Taking determinants proves the assertions. □

If the λ_j are distinct, a right eigenvector w_j satisfies $Z w_j \propto V_\rho e_j$, so the root coordinates are the ratios of the entries indexed by $x_i x_0^{\rho - 1}$ and x_0^ρ. Taking $\beta = x_0$ is always valid here; a generic α separates the roots. No explicit factorization of the u-resultant is needed.

Example 3.7 (Macaulay compression for the running system)**.** For (1), $F_1 = x^2 - x_0^2$, $F_2 = y^2 - xx_0$, and $\rho = 3$. There are ten degree-three monomials and six independent polynomial rows: $x_0F_1, xF_1, yF_1, x_0F_2, xF_2, yF_2$. Choose

$$(b_1, b_2, b_3, b_4) = (x_0^2, x_0x, x_0y, xy), \qquad h = y - \lambda x_0.$$

In degree three use the quotient basis $(x_0^3, x_0^2x, x_0^2y, x_0xy)$. Reduction by $x^2 = x_0^2$ and $y^2 = xx_0$ makes the four rows b_ih equal to

$$H_h = \begin{pmatrix} -\lambda & 0 & 1 & 0 \\ 0 & -\lambda & 0 & 1 \\ 0 & 1 & -\lambda & 0 \\ 1 & 0 & 0 & -\lambda \end{pmatrix}, \qquad \det H_h = \lambda^4 - 1.$$

This is (7) with Z the coefficient-to-quotient map. At a root, $(1, x, y, xy)^T$ lies in its kernel when $\lambda = y$; its entries recover x, y. The constant matrix is the transpose of multiplication by y in the affine basis $(1, x, y, xy)$, reflecting our use of coefficient rows.

Relation to the classical determinant formula. Macaulay expresses the resultant as a quotient of two specified determinants [25, 26]. An arbitrary square minor of a coefficient matrix need not equal the resultant or have exactly its zero set. In particular, one cannot always retain all polynomial rows: for three quadratics, $\rho = 4$ and relations such as $F_2F_1 - F_1F_2 = 0$ already make those rows dependent. The compression above gives a direct, verifiable eigenproblem; classical determinant quotients use their own prescribed row selection and extraneous-factor removal. See Jónsson–Vavasis [22] for a numerical resultant construction.

Reduction and recovery. An $N \times N$ generalized eigenvalue problem $H_\alpha w = \lambda H_\beta w$, obtained from homogeneous polynomial relations and a quotient basis. Under the hypotheses above it is regular and has exactly the projected roots as its eigenvalues.

Numerical aspects. The uncompressed matrix has $\binom{\rho+d}{d}$ columns. Computing its numerical rank, choosing a quotient basis, and scaling polynomial rows are substantial parts of the algorithm. Algebraically equivalent compressions can have different conditioning. The conditioning results for a particular Macaulay pencil [19] therefore need not transfer unchanged to every quotient compression.

3.4 Multiparameter eigenvalue formulations

Idea. Represent each polynomial as a determinant of an affine matrix pencil:

$$p_k(x) = \det W_k(x), \qquad W_k(x) = A_{k,0} + \sum_{i=1}^{d} x_i A_{k,i}, \quad A_{k,i} \in \mathbb{C}^{m_k \times m_k}.$$

Then $P(x) = 0$ if and only if there are nonzero vectors satisfying the *multiparameter eigenvalue problem*

$$W_k(x)v_k = 0, \qquad v_k \neq 0, \qquad k = 1, \ldots, d. \tag{8}$$

The representation trades polynomial degree for matrix size. Explicit constructions are available; a *uniform* representation has entries affine-linear in the polynomial coefficients as well as in the variables, allowing products of one coefficient and one variable. For fixed $d \geq 2$, the smallest size for a representation uniform over all total-degree-at-most-n polynomials is $\Theta(n^{d/2})$; in two variables explicit constructions have size $2n+1$ or $2n-1$ [4]. Choosing a representation is itself an algorithmic step, separate from solving its eigenproblem.

Tensor Cramer's rule. On $\mathbb{C}^{m_1} \otimes \cdots \otimes \mathbb{C}^{m_d}$ define

$$\Delta_0 = \sum_{\sigma \in \mathfrak{S}_d} \operatorname{sgn}(\sigma) A_{1,\sigma(1)} \otimes \cdots \otimes A_{d,\sigma(d)},$$

and define Δ_i by replacing column i of the coefficient array $(A_{k,l})_{k,l=1}^{d}$ with $(-A_{k,0})_k$ before this expansion. For two parameters, writing $W_k = A_k + xB_k + yC_k$, this means

$$\begin{aligned} \Delta_0 &= B_1 \otimes C_2 - C_1 \otimes B_2, \\ \Delta_1 &= C_1 \otimes A_2 - A_1 \otimes C_2, \\ \Delta_2 &= A_1 \otimes B_2 - B_1 \otimes A_2. \end{aligned}$$

If (8) holds and $v = v_1 \otimes \cdots \otimes v_d$, then

$$\Delta_i v = x_i \Delta_0 v, \qquad i = 1, \ldots, d.$$

To check this, substitute $-A_{k,0}v_k = \sum_l x_l A_{k,l} v_k$ into the determinant expansion: every term with a repeated column cancels, leaving only the term $l = i$. For scalar blocks this is ordinary Cramer's rule.

When Δ_0 is invertible, the multiparameter problem is called *nonsingular*. Atkinson's theorem states that $\Gamma_i = \Delta_0^{-1}\Delta_i$ commute and that their joint eigenvalues are exactly the solutions of (8) [2]. This converse is the substantive spectral theorem; the preceding tensor calculation supplies the elementary forward implication. One solves a separating linear combination $\sum_i a_i \Gamma_i$ and recovers all coordinates on its common eigenvectors. If the chosen combination has repeated eigenvalues, its invariant subspaces must be resolved using the remaining operators.

Example 3.8 (A nonsingular two-parameter problem)**.** The running system has the representations

$$W_1 = \begin{pmatrix} x & 1 \\ 1 & x \end{pmatrix}, \qquad W_2 = \begin{pmatrix} y & x \\ 1 & y \end{pmatrix}.$$

Their determinants are $x^2 - 1$ and $y^2 - x$. In the tensor order $(e_1 \otimes e_1, e_1 \otimes e_2, e_2 \otimes e_1, e_2 \otimes e_2)$,

$$\Delta_0 = I_4, \qquad \Delta_1 = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \quad \Delta_2 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{pmatrix}.$$

Direct multiplication gives $\Delta_2^2 = \Delta_1$ and $\Delta_1^2 = I_4$. Moreover $\det(\lambda I_4 - \Delta_2) = \lambda^4 - 1$, so its four eigenvalues are $y = 1, -1, i, -i$; the corresponding x is y^2. At any root one can check the common eigenvector explicitly:

$$v_1 = (1, -x)^T, \quad v_2 = (-y, 1)^T, \quad v = v_1 \otimes v_2 = (-y, 1, xy, -x)^T.$$

Thus $\Delta_1 v = xv$ and $\Delta_2 v = yv$, and $x = -v_4/v_2$, $y = -v_1/v_2$. Solving only for x leaves two-dimensional eigenspaces; y separates all four roots.

Reduction and recovery. Coupled generalized eigenproblems $\Delta_i v = x_i \Delta_0 v$ on a tensor space of dimension $\prod_k m_k$. In the nonsingular case, a separating linear combination gives an ordinary or generalized eigenproblem followed by recovery on common eigenvectors.

Numerical aspects. Invertibility of Δ_0 is a hypothesis to check, not a consequence of having determinantal representations. A singular Δ_0 does not by itself make each pencil singular: a regular pencil can have eigenvalues at infinity. Nor does excess tensor dimension alone distinguish infinite eigenvalues from a singular pencil. For genuinely singular coupled pencils, a structure-aware staircase or related reduction is needed before extracting the finite joint spectrum [33]. For the distinct issue of condition numbers of singular matrix polynomials, see [24]. Representation size, scaling, and the tensor dimension all affect practical accuracy and cost. Backward error and conditioning of the multiparameter problem are studied in [21]; the conditioning of the operator-determinant rootfinding reduction is studied in [19].

4 Family II: scalar eliminants

A scalar eliminant records the values of a coordinate or linear form on the root set. It must be accompanied by a way to recover the other coordinates. The key distinction is whether the chosen scalar value separates the roots.

4.1 Gröbner elimination and the shape lemma

For a coordinate x_d, the elimination ideal $I \cap \mathbb{C}[x_d]$ is generated by a unique monic polynomial f. Proposition 2.1 shows directly that

$$f(T) = \prod_{\eta \in \{(z_j)_d : 1 \le j \le N\}} (T - \eta). \tag{9}$$

Indeed, $h(x_d)$ belongs to I exactly when h vanishes at all these distinct values. Thus f is the *minimal* polynomial of M_{x_d}; it equals the characteristic polynomial only when x_d separates the roots.

To compute f from the equations, choose lexicographic order $x_1 \succ \cdots \succ x_d$. The elimination theorem says that the members involving only x_d of a Gröbner basis form a basis of $I \cap \mathbb{C}[x_d]$ [10, Ch. 3]. The short reason is that division of a univariate member of I must reduce its leading power of x_d; any Gröbner-basis leading monomial dividing this power is itself a power of x_d, and lexicographic order then forces the entire divisor to be univariate. Iterating gives a univariate reduction to zero.

Proposition 4.1 (Shape lemma in the reduced case)**.** *If x_d separates the roots, there are unique polynomials h_i of degree less than N such that*

$$I = \langle x_1 - h_1(x_d), \dots, x_{d-1} - h_{d-1}(x_d), f(x_d) \rangle, \qquad \deg f = N. \tag{10}$$

The displayed generators form the reduced lexicographic Gröbner basis.

Proof. Interpolate $h_i((z_j)_d) = (z_j)_i$ at the N distinct values. Each displayed generator vanishes at all roots and hence belongs to I. The quotient by the displayed ideal is isomorphic to $\mathbb{C}[T]/(f)$ and has dimension N, as does R/I, so the inclusion of ideals is equality. The leading monomials $x_1, \dots, x_{d-1}, x_d^N$ are pairwise coprime; the generators are monic and their other terms are reduced. This proves the Gröbner-basis assertion by Buchberger's criterion. □

This elementary proof explains the usual shape lemma [17, 9]: a separating coordinate makes the whole quotient algebra generated by one element. A generic linear change of coordinates provides such a coordinate, but does not guarantee that the projected roots are well separated numerically.

Example 4.2 (Elimination in the running system)**.** With $x \succ y$, set $a = x - y^2 = -q$. Reducing p by a gives

$$p - (x + y^2)a = y^4 - 1.$$

Conversely, $p = (x + y^2)a + (y^4 - 1)$ and $q = -a$, so

$$I = \langle x - y^2,\ y^4 - 1 \rangle.$$

Solving $y^4 - 1 = 0$ and evaluating $x = y^2$ recovers all four roots. If we eliminate y instead, the eliminant is $x^2 - 1$, of degree two. Its two roots require the additional fiber solves $y^2 = x$; the characteristic polynomial of M_x is $(T^2 - 1)^2$. This is why an eliminant need not have degree N.

Reduction and recovery. Compute a lexicographic Gröbner basis, solve the univariate eliminant, and recover the other coordinates by the shape formulas when the projection separates. Without separation, solve the remaining fibers or change the projection. A univariate companion solve adds its own representation and perturbation model.

Computation and literature. Buchberger's algorithm and the F4/F5 algorithms compute Gröbner bases over exact coefficient fields [5, 14, 15]. With rational input one can combine exact elimination with arbitrary-precision numerical rootfinding. For approximate coefficients, deciding whether a leading coefficient is zero is a rank decision; exact leading-term algorithms can change discontinuously across such decisions. This motivates the numerical normal-form constructions of Section 5.1. Coefficient growth and scalar root conditioning must still be considered even when elimination itself is performed exactly [19].

4.2 Rational univariate representation

Choose a separating linear form $t = u^T x$ and put $t_j = t(z_j)$. The reduced-case rational univariate representation (RUR) consists of

$$f(T) = \prod_{j=1}^{N} (T - t_j), \qquad g_v(T) = \sum_{j=1}^{N} v(z_j) \prod_{k \neq j} (T - t_k), \tag{11}$$

for $v = 1, x_1, \ldots, x_d$. Here $g_1 = f'$ and $\deg g_v < N$.

Proposition 4.3 (Rational recovery and a coefficient construction)**.** *For every root z_j,*

$$(z_j)_i = \frac{g_{x_i}(t_j)}{f'(t_j)}.$$

The polynomials in (11) *can be computed without knowing the roots by*

$$f(T) = \det(TI - M_t), \qquad g_v(T) = \operatorname{trace}\Big(M_v \operatorname{adj}(TI - M_t)\Big). \tag{12}$$

Proof. At $T = t_j$, all terms in g_v except the j-th vanish. The surviving product is $f'(t_j) \neq 0$, proving recovery. To prove (12), simultaneously diagonalize M_t and M_v using (2); the adjugate of the diagonal matrix has j-th entry $\prod_{k \neq j}(T - t_k)$, and trace gives the sum. □

The trace identity is a conceptual recipe. Implementations can obtain the needed coefficients from traces $\operatorname{trace}(M_v M_t^k)$ and Newton identities, rather than forming the full polynomial adjugate [34, 1]. In the radical case, RUR and polynomial shape coordinates are equivalent: f' is invertible modulo f and $h_i \equiv g_{x_i}/f' \pmod f$. The rational form is useful for exact representation and arithmetic; it is not a general numerical conditioning guarantee.

Example 4.4 (Rational recovery in the running system)**.** Take $t = y$. We have $f(T) = T^4 - 1$, $f'(T) = 4T^3$, and

$$g_x(T) \equiv T^2 f'(T) \equiv 4T \pmod f, \qquad g_y(T) \equiv T f'(T) \equiv 4 \pmod f.$$

At a root λ of f the recovered point is

$$x = \frac{4\lambda}{4\lambda^3} = \lambda^2, \qquad y = \frac{4}{4\lambda^3} = \lambda,$$

using $\lambda^4 = 1$. A form need not be a coordinate: $t = x + y$ also separates these roots, taking values $2, 0, -1 + i, -1 - i$. Its polynomial is

$$f_t(T) = T(T - 2)(T^2 + 2T + 2) = T^4 - 2T^2 - 4T.$$

Since $y = t - x$ and $x^2 = 1$, the equation $y^2 = x$ gives $x = (t^2 + 1)/(2t + 1)$, with $y = t - x$; the denominator is nonzero at the four listed values. This is an alternative rational parametrization of exactly the same roots.

Reduction and recovery. Find a separating form t; compute f and the coordinate numerators; solve the scalar degree-N equation $f(T) = 0$; evaluate the rational recovery formulas. If the chosen form does not separate, these reduced-case formulas with denominator f' cannot be used unchanged.

Computation and literature. Rouillier's RUR and Kronecker-style geometric resolution make univariate descriptions effective [34, 18, 23]. An absolutely continuous distribution for the coefficients of t avoids the finitely many collision hyperplanes with probability one. It need not avoid nearly colliding projections, however, and both the scalar solve and rational recovery can then be sensitive. Making f monic fixes a scalar normalization, not the coefficient norm, the scaling of T, or whether perturbations may change its leading coefficient. Those choices remain part of a condition analysis.

5 Family III: normal-form methods

5.1 Constructing the multiplication operators

Proposition 2.2 explains the eigenvalues. The computational task is to obtain $[x_i b_k]_{\mathcal{B}}$ from the coefficients of P. A *normal form* is the linear reduction map taking a polynomial to its unique representative in span $\mathcal{B}$ modulo I. A Gröbner basis supplies one such map: repeatedly replace leading monomials using the polynomial relations until only standard monomials remain. Border bases allow more flexible choices of $\mathcal{B}$, with consistency conditions ensuring that the resulting multiplication operators commute [28, 13].

Example 5.1 (Multiplication matrices for the running system)**.** The relations $x^2 = 1$, $y^2 = x$ reduce every monomial to the span of $\mathcal{B} = (1, x, y, xy)$. Since $\dim A = 4$, these four classes form a basis. Columnwise multiplication gives

$$M_x = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad M_y = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}. \tag{13}$$

For example, the last column of M_y records $y(xy) = xy^2 = x^2 = 1$. Directly, $M_y^2 = M_x$ and $M_y^4 = I$. The vectors $e_1, M_y e_1, M_y^2 e_1, M_y^3 e_1$ form a basis, so M_y is cyclic with minimal and characteristic polynomial $T^4 - 1$. Also, M_x exchanges two pairs of basis vectors. Thus

$$\det(TI - M_y) = T^4 - 1, \qquad \det(TI - M_x) = (T^2 - 1)^2.$$

Diagonalize M_y. If $M_y r = \lambda r$, then $M_x r = \lambda^2 r$, so the corresponding root is (λ^2, λ). This gives all four roots with no matching of independently ordered eigenvalue lists. Diagonalizing M_x alone would leave two two-dimensional eigenspaces that M_y must further resolve.

Normal forms need only be known on a finite polynomial space containing $\mathcal{B}$ and its products by the coordinates. This leads to the following practical formulation.

Proposition 5.2 (A truncated normal form)**.** *Let U be a finite-dimensional polynomial space containing $\mathcal{B}$ and all $x_i\mathcal{B}$. Suppose a linear map $L : U \to \mathbb{C}^N$ satisfies $\ker L = I \cap U$, and the classes of $\mathcal{B}$ form a basis of A. Put*

$$L_{\mathcal{B}} = (L(b_1) \ \cdots \ L(b_N)), \qquad L_{x_i\mathcal{B}} = (L(x_i b_1) \ \cdots \ L(x_i b_N)).$$

Then $L_{\mathcal{B}}$ is invertible and

$$M_{x_i} = L_{\mathcal{B}}^{-1} L_{x_i\mathcal{B}}. \tag{14}$$

Proof. A linear combination of $\mathcal{B}$ annihilated by L belongs to I, so its coefficients vanish. This proves invertibility. Each column of the right side of (14) represents the unique combination of $\mathcal{B}$ whose difference from $x_i b_k$ is annihilated by L, hence belongs to I. □

One constructs L from linear relations between monomials: a sufficiently large Macaulay matrix lists multiples of the input equations as rows. The quotient by its row space, equivalently the cokernel of its transpose, gives the required quotient map when the rank and degree conditions hold. The word "sufficiently" is essential; too small a truncation can miss relations. QR with pivoting or an SVD can then select $\mathcal{B}$ for which $L_{\mathcal{B}}$ is well conditioned. This is the approach of stabilized and truncated normal forms [41, 40, 29]. It explains both the flexibility of these methods and their connection to Section 3.3.

Reduction and recovery. Compute multiplication matrices in a chosen quotient basis. Choose a separating $g = \sum_i \gamma_i x_i$ and solve the ordinary eigenproblem for $M_g = \sum_i \gamma_i M_{x_i}$. Recover each coordinate from the action of the other matrices on the same eigenspace. Equivalently, use evaluation eigenvectors of M_g^T when the basis contains appropriate coordinate monomials.

Numerical aspects and literature. Multiplication matrices need not be normal, although the running example happens to give permutation matrices. A Schur-based implementation can work with clusters or invariant subspaces instead of individual eigenvectors [8]. The cost of a dense N-dimensional eigensolve is $O(N^3)$; constructing a reliable quotient can itself be substantial. Foundational numerical treatments are Auzinger–Stetter, Möller–Stetter, and Stetter [3, 27, 35, 36]. For modern variants and the effects of clustering, see [40, 32]. No basis-selection rule makes intrinsically ill-conditioned roots disappear, and improving a quotient solve does not by itself bound the conditioning of its eigenvectors.

6 Conditioning: specify what is perturbed

The constructions above are exact algebraic statements. To compare their numerical behavior, one must specify an input norm, an output norm, and which entries or coefficients may change. This section gives the basic formulas rather than an average-case classification.

6.1 Equation values, coefficients, and representations

At a simple root z, differentiation gives

$$\delta z = -J(z)^{-1}\delta P(z). \tag{15}$$

Thus $\|J^{-1}\|_2$ is the absolute condition for Euclidean perturbations of the *equation values*. Suppose instead that all equations use a fixed basis $\phi = (\phi_1, \ldots, \phi_m)^T$, with $p_i = c_i^T\phi$, and the coefficient array has its Euclidean norm.

Proposition 6.1 (Coefficient condition of a root)**.** *With $S(z) = \sum_k |\phi_k(z)|^2$,*

$$\kappa_{\text{coeff}}(P, z) = \sqrt{S(z)}\,\|J(z)^{-1}\|_2.$$

For a scalar output $g(z)$, replace $\|J^{-1}\|_2$ by $\|Dg(z)J^{-1}\|_2$.

Proof. The coefficient evaluation map $E_z : \delta c \mapsto \delta P(z)$ satisfies $E_z E_z^* = S(z)I_d$. Therefore the squared operator norm of $J^{-1}E_z$ is $S(z)\|J^{-1}\|_2^2$. Apply the chain rule for the scalar output. □

For example, with $p(T) = \sum_{k=0}^m a_k T^k$, the coefficient condition of a simple root λ is

$$\frac{\left(\sum_{k=0}^m |\lambda|^{2k}\right)^{1/2}}{|p'(\lambda)|},$$

when all coefficients are perturbable. If a monic leading coefficient is held fixed, the sum ends at $m-1$. The value-perturbation expression $1/|p'|$ omits this evaluation norm. With a sup norm on a fixed unit disk, evaluation outside the disk also needs care: a degree-m perturbation bounded by ϵ on the disk can reach $\epsilon|\lambda|^m$ at an exterior point.

For nonzero data and output, a normwise relative condition multiplies the corresponding absolute condition by the input norm and divides by the output norm. This does not change which perturbations are allowed. It also does not remove dependence on a nonunitary change of basis.

6.2 Eigenvectors, interpolation, and basis choice

For a simple eigenvalue λ of a matrix M, choose right and left eigenvectors r, w, so $Mr = \lambda r$ and $w^*M = \lambda w^*$. Differentiating and multiplying by w^* gives

$$\delta\lambda = \frac{w^*(\delta M)r}{w^*r}, \qquad \kappa_{\text{eig}} = \frac{\|w\|_2\|r\|_2}{|w^*r|},$$

for absolute spectral-norm matrix perturbations; a rank-one perturbation attains the bound. For M_g with g separating, Proposition 2.2 therefore gives

$$\kappa_{\text{eig}}(g(z_j), M_g) = \Lambda_j := \|V^{-T}e_j\|_2\|Ve_j\|_2 \geq 1. \tag{16}$$

The normalization is $v_j^T V^{-T} e_j = 1$; in the Hermitian convention the left eigenvector is $\overline{v_j}$.

The inverse Gram matrix makes the geometry explicit:

$$\Lambda_j^2 = (V^*V)_{jj}\Big((V^*V)^{-1}\Big)_{jj} = \frac{\|v_j\|_2^2}{\operatorname{dist}(v_j, \operatorname{span}\{v_k : k \neq j\})^2}.$$

To see the last equality, project v_j orthogonally onto the complement of the other columns. The vector dual to v_j is this residual divided by its squared norm. Thus conditioning measures an interpolation angle involving all the evaluation vectors, not only the distance to the nearest root.

Example 6.2 (A basis can create poor eigenvalue conditioning)**.** For $p = x^2 - 1$, the basis $(1, x)$ gives $M_x = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, whose two eigenvalues have condition one. In the basis $(1, \epsilon x)$, $\epsilon > 0$,

$$M_x = \begin{pmatrix} 0 & \epsilon \\ \epsilon^{-1} & 0 \end{pmatrix}.$$

At the root 1 the evaluation vector is $(1, \epsilon)^T$ and the idempotent $(1 + x)/2$ has coefficient vector $(1/2, 1/(2\epsilon))^T$. Hence

$$\Lambda = \frac{\epsilon + \epsilon^{-1}}{2}.$$

The roots remain ± 1 and $|p'(\pm 1)| = 2$, but the unstructured eigenvalue condition grows without bound as $\epsilon \to 0$. This is a representation effect.

6.3 Matrix polynomials and scalar eliminants

For a regular matrix polynomial $R(\lambda) = \sum_{k=0}^{m} R_k \lambda^k$ and a simple finite eigenvalue, let $R(\lambda)v = 0$ and $w^*R(\lambda) = 0$. The same differentiation gives

$$\delta\lambda = -\frac{w^* \delta R(\lambda) v}{w^* R'(\lambda) v}.$$

Under the coefficient norm $(\sum_k \|\delta R_k\|_F^2)^{1/2}$, its condition is

$$\frac{\left(\sum_{k=0}^{m} |\lambda|^{2k}\right)^{1/2} \|w\|_2 \|v\|_2}{|w^* R'(\lambda) v|}. \tag{17}$$

Cauchy–Schwarz proves the upper bound, and coefficientwise aligned rank-one perturbations attain it. Other coefficient weights or function norms give different evaluation factors [42]. For the pencil (M, I), allowing independent perturbations of both matrices with their maximum spectral norm gives $\Lambda(1 + |\lambda|)$; holding I fixed gives (16).

For a monic scalar eliminant with distinct roots t_j,

$$f'(t_j) = \prod_{k \neq j} (t_j - t_k).$$

Taking logarithms expresses its value-perturbation sensitivity through pairwise projected distances. This elementary identity does not by itself compare coefficient conditioning across methods: the evaluation factors, eliminant coefficient norm, and coordinate-recovery step still enter. A projection can bring distinct, well-separated multivariate roots close together.

Finally, if $R(c, \lambda)$ is built exactly from the input coefficients, then a coefficient perturbation induces $\delta R = D_c R[\delta c]$, a structured subset of all matrix-polynomial perturbations. The chain rule must recover (15) for the exact composite solve and recovery map. A backward-stable QR or QZ computation controls perturbations of the matrix or pencil it receives; translating them back to the original polynomial coefficients requires additional analysis of construction, linearization, and recovery.

Noferini–Townsend and Graf–Townsend exhibit systems for which intermediate subproblems are much more sensitive than the original roots under their specified perturbation models [31, 19]. These are worst-case results, not statements that every input is unstable. Nor do the exact identities in this survey establish favorable average-case behavior. They identify the quantities and choices that a probabilistic or smoothed analysis must control.

7 Synthesis and comparison

7.1 What the representations have in common

The quotient algebra gives a precise comparison. For a separating t, the eliminant is $\det(TI - M_t)$; without separation, its squarefree minimal polynomial has smaller degree. Under the stated regularity and no-extraneous-root assumptions, a hidden-variable resultant determinant represents the same projected root values. Macaulay quotient compression and truncated normal forms compute different coordinates for the same finite-dimensional multiplication maps.

One should therefore distinguish equality of the represented zero set, equality of scalar determinants including multiplicities and factors, and similarity or equivalence of matrix representations. The first does not imply the latter two, and none alone implies equality of numerical condition numbers.

7.2 An optional duality viewpoint: residues

There is a useful explanation of the Jacobian factors that appear with Bézoutians. For a reduced complete intersection, define

$$\tau(f) = \operatorname{trace} M_f = \sum_j f(z_j), \qquad \mathcal{R}(f) = \sum_j \frac{f(z_j)}{\det J(z_j)}.$$

The latter is the global residue functional in the simple-root case [6]. The pairings $\tau(fg)$ and $\mathcal{R}(fg)$ are nondegenerate, but they are different: if $\mathcal{J} = [\det DP] \in A$, then $\tau(f) = \mathcal{R}(\mathcal{J}f)$. Nondegeneracy follows immediately in the idempotent basis, where both pairings are diagonal with nonzero diagonal entries.

Proposition 7.1 (Univariate Bézoutian and residue duality)**.** *Let p be monic and squarefree of degree m. Then $B(p,1) = H^{-1}$, where H is the residue Gram matrix in the monomial basis of $\mathbb{C}[x]/(p)$:*

$$H_{ab} = \sum_{p(\lambda)=0} \frac{\lambda^{a+b}}{p'(\lambda)}, \qquad 0 \le a, b < m,$$

Proof. For each root λ_j, the Bézoutian evaluated at its second argument is $p(s)/(s-\lambda_j) = p'(\lambda_j)\ell_j(s)$. Thus $B(p,1)V = V^{-T}\operatorname{diag}(p'(\lambda_j))$, whereas $H = V\operatorname{diag}(1/p'(\lambda_j))V^T$. Multiplication gives $B(p,1)H = I$. □

For $p = x^2 - 1$, this residue Gram matrix and its inverse are both $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$; the ordinary trace Gram matrix is $2I$. Calling Bézoutian duality simply trace duality would miss this Jacobian weight.

More generally, if $P(x) = Q(x)(x - z_j)$, then

$$[\det Q] = \det J(z_j)[\ell_j] \quad \text{in } A.$$

At another root z_k, $Q(z_k)(z_k - z_j) = 0$ makes its determinant vanish; at z_j, differentiation gives $Q(z_j) = J(z_j)$. Evaluation proves the identity. It explains why determinant numerators in normal-form condition formulas can equally be written as norms of Lagrange idempotents [19].

7.3 A comparison to use when choosing a construction

For exact input and symbolic output, elimination and RUR provide explicit root descriptions. For numerical work, retaining matrix structure can avoid forming large scalar eliminant coefficients, but introduces basis, linearization, and eigenvector issues. Normal forms make basis selection explicit; resultants make the relation to coefficient elimination explicit; multiparameter formulations make affine matrix pencils explicit. The right choice depends on the polynomial support, available arithmetic, desired output, and the conditioning of the construction.

The running example gives a concrete cross-check: Sylvester yields $y^4 - 1$, Bézout yields $1 - y^4$, Gröbner elimination gives $x = y^2$ and $y^4 = 1$, RUR gives the same coordinates rationally, and multiplication by y has characteristic polynomial $T^4 - 1$. The Macaulay and multiparameter examples recover the same four points through pencils. Their agreement is an algebraic fact; their response to floating-point perturbations depends on which of these representations is used.

Method	Intermediate problem and size	Conditions and recovery
Sylvester	For two generic box-degree-n equations: a $2n \times 2n$ PEP of grade n.	Distinct projected roots and no common fiber root at infinity; recover the eliminated coordinate from a null vector or a fiber solve.
Cayley–Bézout	For d generic box-degree-n equations: size $(d-1)!n^{d-1}$, grade dn.	Regularity and the stated multihomogeneous model; recover coordinates from tensor evaluation vectors.
Macaulay / u-resultant	Dense degree-ρ relation matrix, followed here by an $N \times N$ compressed pencil.	Complete intersection with no ordinary projective root at infinity; valid quotient ranks, a nonvanishing denominator, and a separating ratio α/β for individual eigenvector recovery.
Multiparameter	Operator-determinant pencils on a tensor space of dimension $\prod_k m_k$.	The nonsingular theory assumes Δ_0 invertible. Singular formulations need an appropriate reduction; coordinates belong to common eigenspaces.
Gröbner elimination	A scalar eliminant of degree at most N, plus extension equations.	A separating coordinate gives degree N and polynomial shape recovery; otherwise solve the remaining fibers.
RUR	A scalar degree-N polynomial and d rational coordinate formulas.	A separating linear form and nonvanishing recovery denominator; construction is symbolic or based on quotient traces.
Normal form	An $N \times N$ ordinary multiplication eigenproblem.	A valid quotient basis or truncation and a separating multiplier; recover coordinates on shared eigenspaces.

Table 1: The seven constructions. A PEP is a polynomial eigenvalue problem. Sizes in the first two rows refer to full box supports; the Macaulay row instead uses total degrees and $\rho = \sum_i (e_i - 1) + 1$. A regular size-$m$ PEP of grade r has mr eigenvalues counting infinity and algebraic multiplicity; a companion linearization has size mr. These counts are not guarantees of distinct affine roots.

References

[1] María Emilia Alonso, Eberhard Becker, Marie-Françoise Roy, and Thorsten Wörmann. Zeros, multiplicities, and idempotents for zero-dimensional systems. In *Algorithms in Algebraic Geometry and Applications*, volume 143 of *Progress in Mathematics*, pages 1–15. Birkhäuser, Basel, 1996.

[2] Frederick V. Atkinson. *Multiparameter Eigenvalue Problems. Volume I: Matrices and Compact Operators*, volume 82 of *Mathematics in Science and Engineering*. Academic Press, New York, 1972.

[3] Winfried Auzinger and Hans J. Stetter. An elimination algorithm for the computation of all zeros of a system of multivariate polynomial equations. In *Numerical Mathematics, Singapore 1988*, volume 86 of *International Series of Numerical Mathematics*, pages 11–30. Birkhäuser, Basel, 1988.

[4] Ada Boralevi, Jasper van Doornmalen, Jan Draisma, Michiel E. Hochstenbach, and Bor Plestenjak. Uniform determinantal representations. *SIAM Journal on Applied Algebra and Geometry*, 1(1):415–441, 2017.

[5] Bruno Buchberger. An algorithm for finding the basis elements of the residue class ring of a zero dimensional polynomial ideal. *Journal of Symbolic Computation*, 41(3-4):475–511, 2006. Translation of the 1965 PhD thesis (University of Innsbruck) by M. P. Abramson.

[6] Eduardo Cattani and Alicia Dickenstein. Introduction to residues and resultants. In Alicia Dickenstein and Ioannis Z. Emiris, editors, *Solving Polynomial Equations: Foundations, Algorithms, and Applications*, volume 14 of *Algorithms and Computation in Mathematics*, pages 1–61. Springer, Berlin, 2005.

[7] Arthur Cayley. Note sur la méthode d'élimination de Bezout. *Journal für die reine und angewandte Mathematik*, 53:366–367, 1857.

[8] Robert M. Corless, Patrizia M. Gianni, and Barry M. Trager. A reordered Schur factorization method for zero-dimensional polynomial systems with multiple roots. In *Proceedings of the 1997 International Symposium on Symbolic and Algebraic Computation (ISSAC '97)*, pages 133–140, New York, 1997. ACM.

[9] David A. Cox, John Little, and Donal O'Shea. *Using Algebraic Geometry*, volume 185 of *Graduate Texts in Mathematics*. Springer, New York, second edition, 2005.

[10] David A. Cox, John Little, and Donal O'Shea. *Ideals, Varieties, and Algorithms*. Undergraduate Texts in Mathematics. Springer, Cham, fourth edition, 2015.

[11] Alicia Dickenstein and Ioannis Z. Emiris. Multihomogeneous resultant formulae by means of complexes. *Journal of Symbolic Computation*, 36(3–4):317–342, 2003.

[12] Arthur L. Dixon. The eliminant of three quantics in two independent variables. *Proceedings of the London Mathematical Society (2)*, 7:49–69, 1909.

[13] Mohamed Elkadi and Bernard Mourrain. *Introduction à la résolution des systèmes polynomiaux*, volume 59 of *Mathématiques et Applications*. Springer, Berlin, 2007.

[14] Jean-Charles Faugère. A new efficient algorithm for computing Gröbner bases (F4). *Journal of Pure and Applied Algebra*, 139(1-3):61–88, 1999.

[15] Jean-Charles Faugère. A new efficient algorithm for computing Gröbner bases without reduction to zero (F5). In *Proceedings of the 2002 International Symposium on Symbolic and Algebraic Computation (ISSAC '02)*, pages 75–83, New York, 2002. ACM.

[16] Israel M. Gelfand, Mikhail M. Kapranov, and Andrei V. Zelevinsky. *Discriminants, Resultants, and Multidimensional Determinants*. Mathematics: Theory & Applications. Birkhäuser, Boston, 1994.

[17] Patrizia Gianni and Teo Mora. Algebraic solution of systems of polynomial equations using Groebner bases. In *Applied Algebra, Algebraic Algorithms and Error-Correcting Codes (AAECC-5)*, Lecture Notes in Computer Science, pages 247–257, Berlin, 1989. Springer.

[18] Marc Giusti, Grégoire Lecerf, and Bruno Salvy. A Gröbner free alternative for polynomial system solving. *Journal of Complexity*, 17(1):154–211, 2001.

[19] Emil Graf and Alex Townsend. Numerical instability of algebraic rootfinders. *Linear Algebra and its Applications*, 729:308–336, 2026.

[20] Uwe Helmke and Paul A. Fuhrmann. Bezoutians. *Linear Algebra and its Applications*, 122–124:1039–1097, 1989.

[21] Michiel E. Hochstenbach and Bor Plestenjak. Backward error, condition numbers, and pseudospectra for the multiparameter eigenvalue problem. *Linear Algebra and its Applications*, 375:63–81, 2003.

[22] Guðbjörn F. Jónsson and Stephen A. Vavasis. Accurate solution of polynomial equations using Macaulay resultant matrices. *Mathematics of Computation*, 74(249):221–262, 2005.

[23] Leopold Kronecker. Grundzüge einer arithmetischen Theorie der algebraischen Grössen. *Journal für die reine und angewandte Mathematik*, 92:1–122, 1882.

[24] Martin Lotz and Vanni Noferini. Wilkinson's bus: Weak condition numbers, with an application to singular polynomial eigenproblems. *Foundations of Computational Mathematics*, 20:1439–1473, 2020.

[25] Francis S. Macaulay. Some formulae in elimination. *Proceedings of the London Mathematical Society*, 35:3–27, 1902.

[26] Francis S. Macaulay. *The Algebraic Theory of Modular Systems*, volume 19 of *Cambridge Tracts in Mathematics and Mathematical Physics*. Cambridge University Press, Cambridge, 1916.

[27] H. Michael Möller and Hans J. Stetter. Multivariate polynomial equations with multiple zeros solved by matrix eigenproblems. *Numerische Mathematik*, 70(3):311–329, 1995.

[28] Bernard Mourrain. A new criterion for normal form algorithms. In *Applied Algebra, Algebraic Algorithms and Error-Correcting Codes (AAECC-13)*, volume 1719 of *Lecture Notes in Computer Science*, pages 430–443, Berlin, 1999. Springer.

[29] Bernard Mourrain, Simon Telen, and Marc Van Barel. Truncated normal forms for solving polynomial systems: generalized and efficient algorithms. *Journal of Symbolic Computation*, 102:63–85, 2021.

[30] Yuji Nakatsukasa, Vanni Noferini, and Alex Townsend. Computing the common zeros of two bivariate functions via Bézout resultants. *Numerische Mathematik*, 129(1):181–209, 2015.

[31] Vanni Noferini and Alex Townsend. Numerical instability of resultant methods for multidimensional rootfinding. *SIAM Journal on Numerical Analysis*, 54(2):719–743, 2016.

[32] Suzanna Parkinson, Hayden Ringer, Kate Wall, Erik Parkinson, Lukas Erekson, Daniel Christensen, and Tyler J. Jarvis. Analysis of normal-form algorithms for solving systems of polynomial equations. *Journal of Computational and Applied Mathematics*, 411:114235, 2022.

[33] Bor Plestenjak and Michiel E. Hochstenbach. Roots of bivariate polynomial systems via determinantal representations. *SIAM Journal on Scientific Computing*, 38(2):A765–A788, 2016.

[34] Fabrice Rouillier. Solving zero-dimensional systems through the rational univariate representation. *Applicable Algebra in Engineering, Communication and Computing*, 9(5):433–461, 1999.

[35] Hans J. Stetter. Matrix eigenproblems are at the heart of polynomial system solving. *SIGSAM Bulletin*, 30(4):22–25, 1996.

[36] Hans J. Stetter. *Numerical Polynomial Algebra*. SIAM, Philadelphia, 2004.

[37] Bernd Sturmfels. *Solving Systems of Polynomial Equations*, volume 97 of *CBMS Regional Conference Series in Mathematics*. American Mathematical Society, Providence, RI, 2002.

[38] James Joseph Sylvester. A method of determining by mere inspection the derivatives from two equations of any degree. *Philosophical Magazine*, 16:132–135, 1840.

[39] James Joseph Sylvester. On a theory of the syzygetic relations of two rational integral functions, comprised in the theory of Sturm's functions, and that of the greatest algebraical common measure. *Philosophical Transactions of the Royal Society of London*, 143:407–548, 1853.

[40] Simon Telen, Bernard Mourrain, and Marc Van Barel. Solving polynomial systems via truncated normal forms. *SIAM Journal on Matrix Analysis and Applications*, 39(3):1421–1447, 2018.

[41] Simon Telen and Marc Van Barel. A stabilized normal form algorithm for generic systems of polynomial equations. *Journal of Computational and Applied Mathematics*, 342:119–132, 2018.

[42] Françoise Tisseur. Backward error and condition of polynomial eigenvalue problems. *Linear Algebra and its Applications*, 309:339–361, 2000.

[43] Bartel L. van der Waerden. *Modern Algebra, Volume II*. Frederick Ungar, New York, 1950.